

InfraPulse | iPhone Pro Max capture

LiDAR PLY merge | LA-weighted LLM | no Pi kit

Capture is iPhone Pro / Pro Max only. Export ASCII PLY. A second person scanning the same GPS cell densifies merged.ply. Gyro/compass align walks; the LLM is trained with Los Angeles distress priors.

1. CORE (you already own this)

Got	Qty	Part	Why	Approx
[]	1	iPhone 12 Pro or newer (Pro Max preferred)	LiDAR + gyro + GPS + baro + ARKit	owned
[]	1	3D Scanner App, Polycam, or ARKit export	ASCII PLY per walk	\$0-40
[]	1	Second walker on the same block	Densifies merged.ply	\$0
[]	1	Laptop	Fusion, LLM, dashboard	owned

2. RECOMMENDED

Got	Qty	Part	Why	Approx
[]	1	Phone lanyard / two-hand grip	Higher gyro pose quality	\$10
[]	1	Notes template (heat, trench, rut, tree)	Feeds LA LLM priors	\$0
[]	1	USB-C cable + laptop folder	Drop PLY into data/scenes	\$0

3. DO NOT BUY

Skip	Reason
Raspberry Pi / MPU-6050 kit	Hardware path retired
HC-SR04 ultrasonic	Not NDT
GPR / professional ultrasound	Later handoff, not this MVP
Drones this weekend	No safe flight + merge pipeline

On-device sensors used

LiDAR mesh, camera, gyroscope, accelerometer, magnetometer, GPS, barometer, ARKit tracking state. Files: contrib_*.ply + merged.ply + scene.ipulse.json.